Optimal Control for Powered-Lift UAV Landing

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Powered-Lift UAVs



Arace Griffin Pro

Used for delivery, agricultural spraying and search & rescue — mostly flown autonomously.

1 Medium-scale UAVs

2–7 m Wingspan

2 VTOL

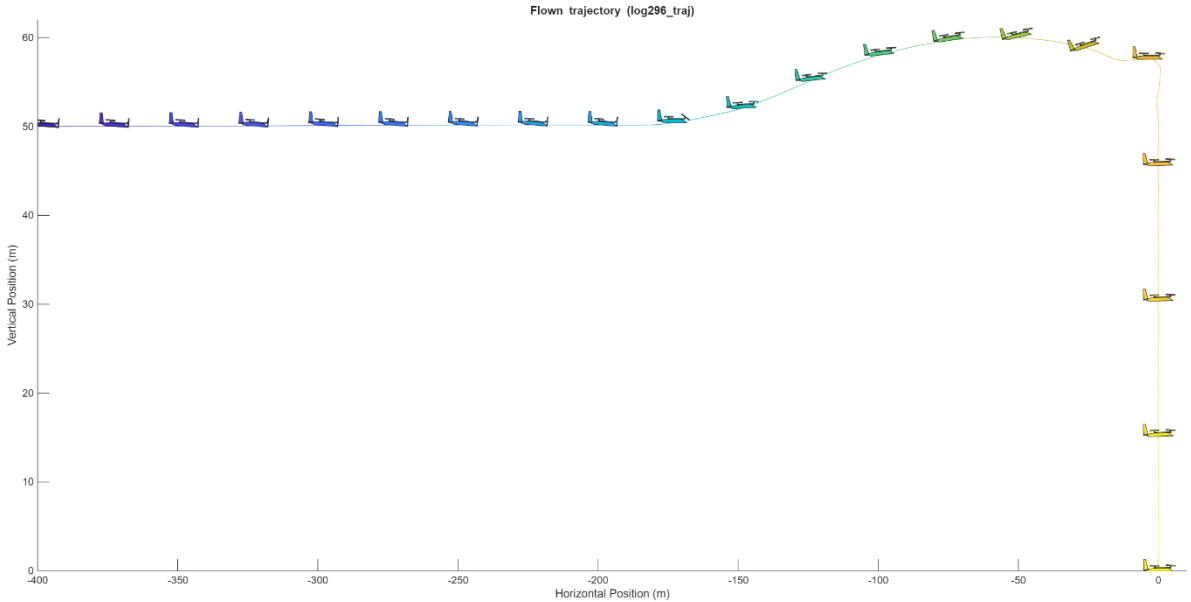
Takes off and lands without a runway

3 Fixed-wing flight

Efficiency and endurance of a plane

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ArduPilot – Stock Tilt-Rotor Landing



90 s

of the 110 s landing is spent hovering down

≈ 3×

more power to hover than to fly as a fixed wing

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Project Goal

1

Efficient

Reduce power consumption during landing using a trajectory-optimisation approach

2

Accessible

Integrated into ArduPilot — no hardware changes required



Arace Griffin Pro

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Workflow

1

Aircraft Dynamical Model

Longitudinal dynamics incl. post-stall aerodynamics and rotor forces

2

Trajectory Optimisation

Minimum-energy landing path within the aircraft's limits

3

Control

Time-varying LQR tracking the trajectory in an ArduPilot Lua script

4

Simulation

Tested in ArduPilot's SITL simulation environment

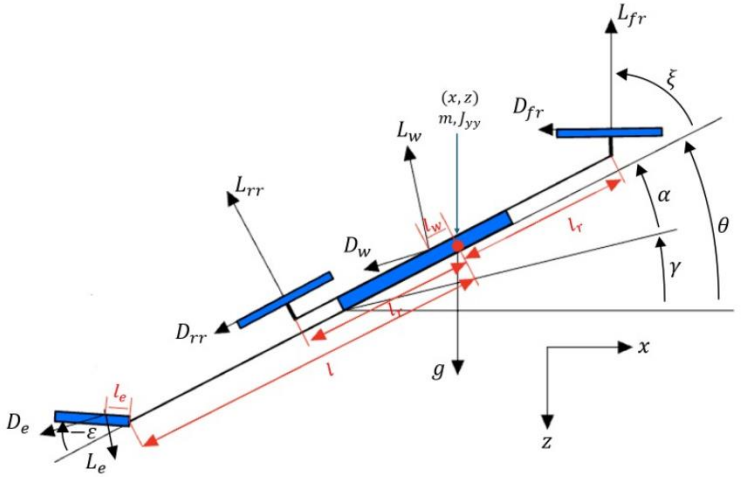
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Aircraft Dynamical Model

$$\dot{X}(t) = f(X(t), U(t), t)$$

$$X = \begin{bmatrix} x & z & \theta & \dot{x} & \dot{x} & \dot{\theta} \end{bmatrix}^T$$

$$U = \begin{bmatrix} \omega_f & \omega_r & \xi & \epsilon \end{bmatrix}^T$$

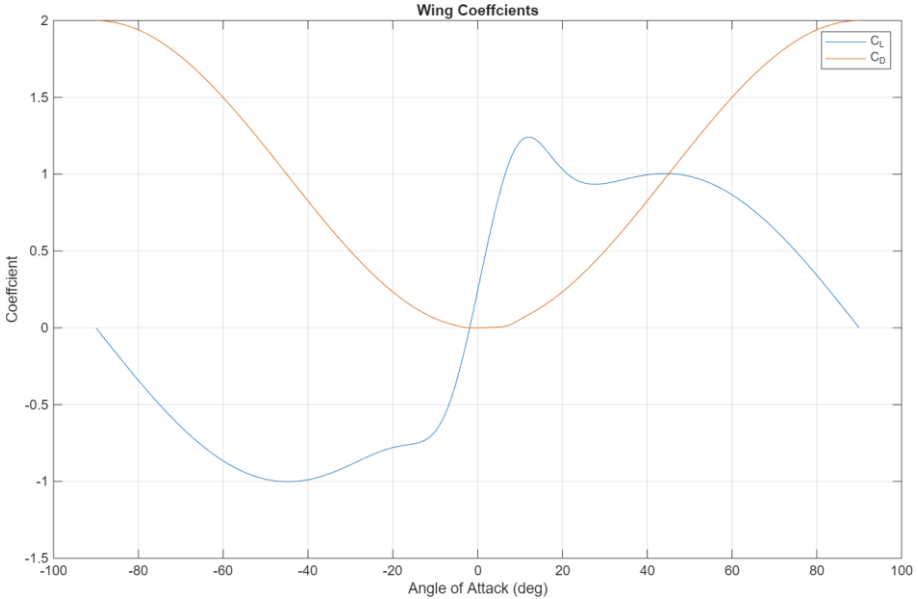


$$\begin{bmatrix} \ddot{x} \\ \ddot{z} \end{bmatrix} = \frac{1}{m}(R(\theta - \alpha)(F_w + F_e)) + R(\theta + \xi)F_{fr} + R(\theta)F_{rr} + mg)$$

$$\ddot{\theta} = \frac{1}{J_{yy}}(l_w R(-\alpha) + F_{w_\perp} + (l\cos(\epsilon) + l_e)R(-\alpha)F_{e_\perp} - l_r R(\theta + \xi)F_{fr_\perp} + l_r R(\theta)F_{rr_\perp})$$

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Aircraft Model – Aero Surfaces



$$L = \frac{1}{2}\rho s|v^2|C_l(\alpha)$$

$$D = \frac{1}{2}\rho s|v^2|C_d(\alpha)$$

$$C_l(\alpha) = (1-\sigma)C_{l_{airfoil}}(\alpha) + \sigma C_{l_{flatplate}}(\alpha)$$

$$C_d(\alpha) = (1-\sigma)C_{l_{airfoil}}(\alpha) + \sigma C_{l_{flatplate}}(\alpha)$$

$$C_{l_{flateplate}}(\alpha) = 2sin(\alpha)cos(\alpha)$$

$$C_{d_{flateplate}}(\alpha) = 2sin^2(\alpha)$$

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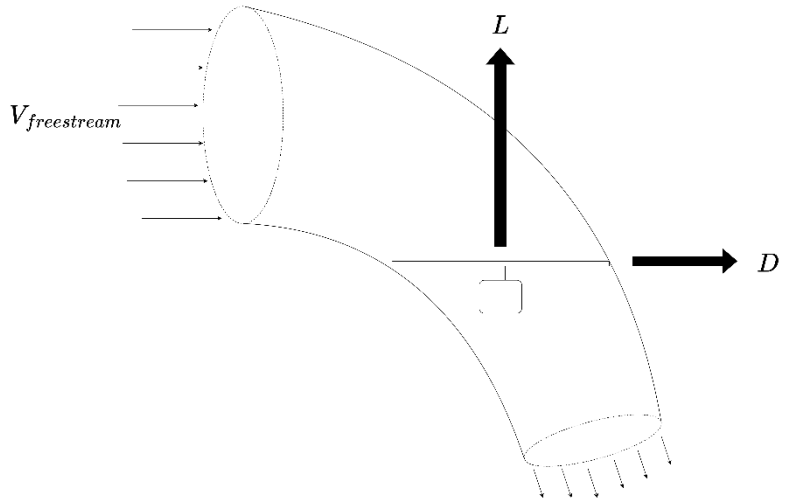
Aircraft Model – Rotors

$$L_{front} = \frac{1}{2} N_b \rho c R_r^3 \omega_{front}^2 C_T, \qquad D_{front} = \frac{L_{front}(\dot{x}_{front} + \dot{y}_{front})}{R_r \omega_{front}}$$

$$L_{rear} = \frac{1}{2} N_b \rho c R_r^3 \omega_{rear}^2 C_T, \qquad D_{rear} = \frac{L_{rear}(\dot{x}_{rear} + \dot{y}_{rear})}{R_r \omega_{rear}}$$

$$\dot{X}_{front} = R_{IR}(\phi, \theta + \xi - \frac{\pi}{2}, \psi), \qquad \dot{X}_{rear} = R_{IR}(\phi, \theta + \xi, \psi), \qquad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$C_T = \frac{2mg}{N_b \rho c R_r^3 (\omega_{front} + \omega_{rear})}$$



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Trajectory Optimisation

$$min \quad J_{trajectory} = \sum_{i=0}^{\frac{t_f}{\Delta t}} 2(P_{front_i} + P_{rear_i})\Delta t$$

$$subject\,to:$$

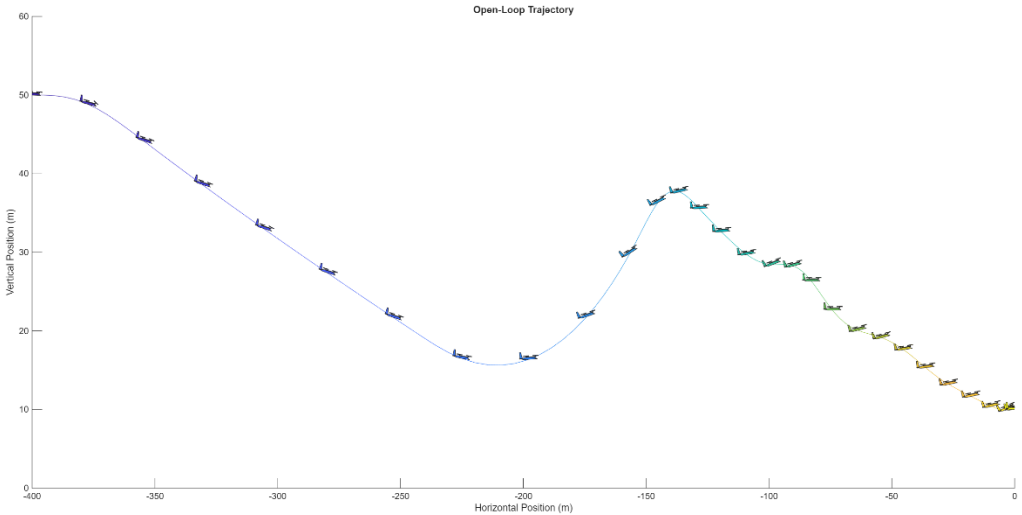
$$X[k+1] = X[k] + f(X[k], U[k])\Delta t$$

$$U_{min} \leq U[k] \leq U_{max}$$

$$X_{final} = X[t_f]$$

$$X_{min} \leq X[k] \leq X_{max}$$

$$X_{init} = X[0]$$



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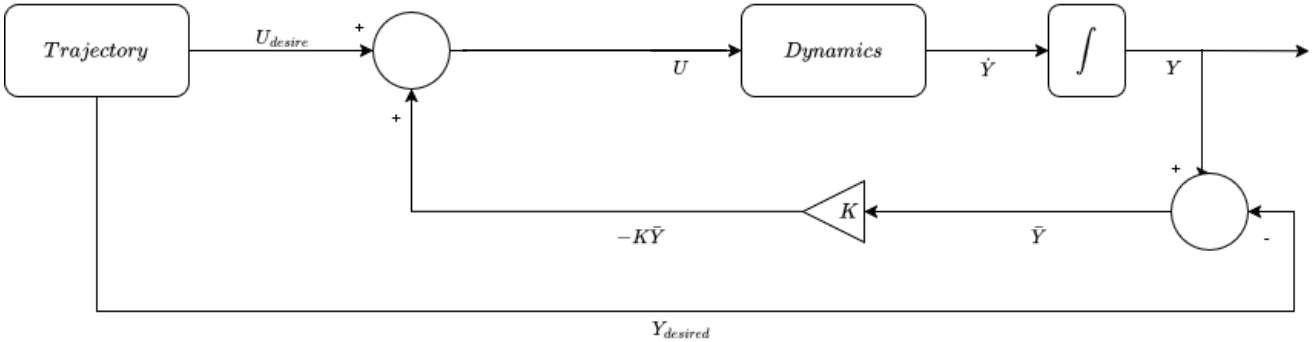
Control - LQR

$$min \quad J_{LQR} = \int_0^{\frac{t_f}{\Delta t}} \bar{X}(t)^T Q \bar{X}(t) + \bar{U}(t)^T R \bar{U}(t)\, dt$$

$$A(t) = \frac{\partial f(t)}{\partial X(t)}, \quad B(t) = \frac{\partial f(t)}{\partial U(t)}$$

$$-\dot{S}(t) = A^T(t)S(t) + S(t)A(t) - S(t)B(t)R^{-1}B^T(t)S(t) + Q$$

$$K(t) = R^{-1}B^T(t)S(t)$$



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ArduPilot Implementation

1 **Lua script**

Runs on existing flight-controller hardware

2 **Trajectory imported**

Pre-computed optimal path loaded for the approach

4 **Lateral control reimplemented**

PD loops keep roll and yaw stable at low airspeed



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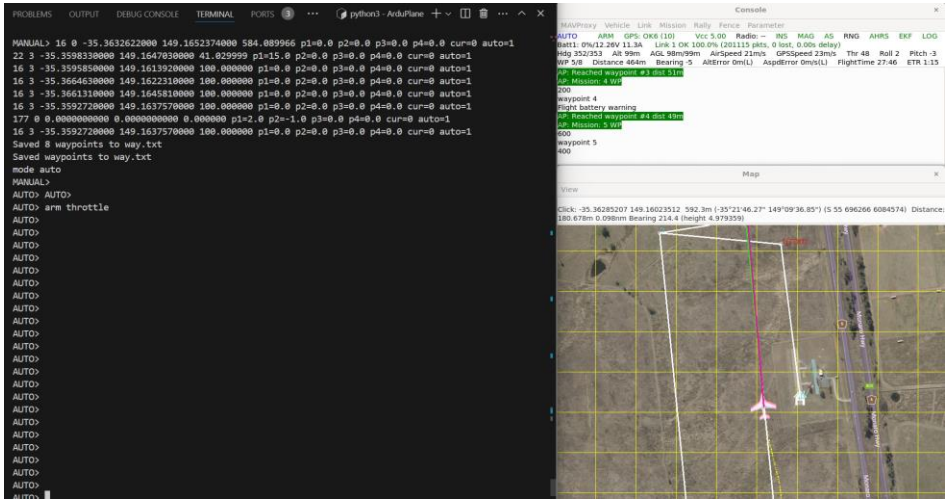
Simulation

1 ArduPilot Simulator

Software-in-the-loop (SITL): the real ArduPilot code and Lua controller fly a simulated aircraft

2 Revised Rotor Model

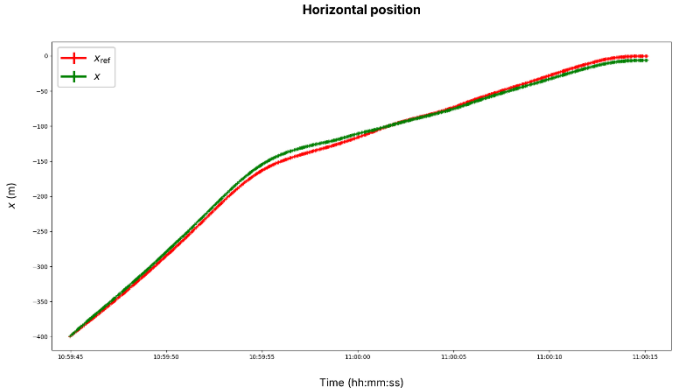
Simulator's rotor model updated to match the aircraft model used for optimisation

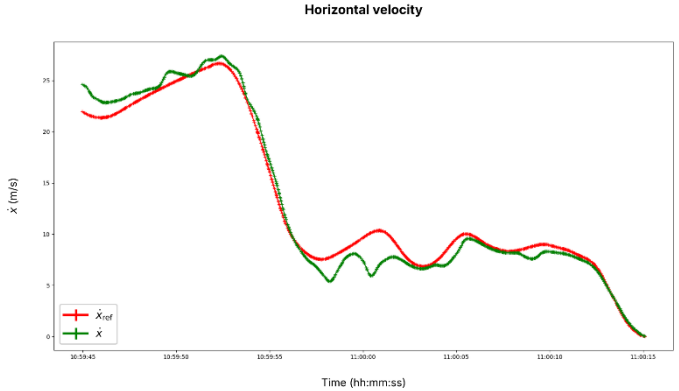


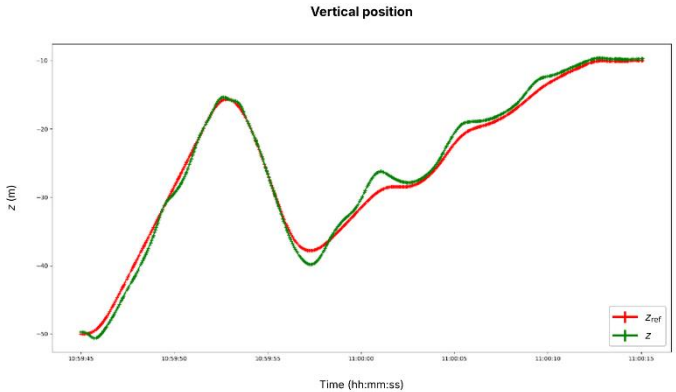
ArduPilot SITL flying a plane mission in MAVProxy. Image: ArduPilot Dev Team, CC BY-SA 3.0

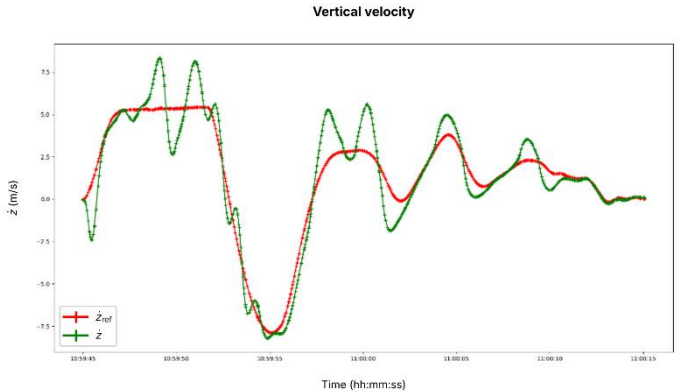
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Control – Results



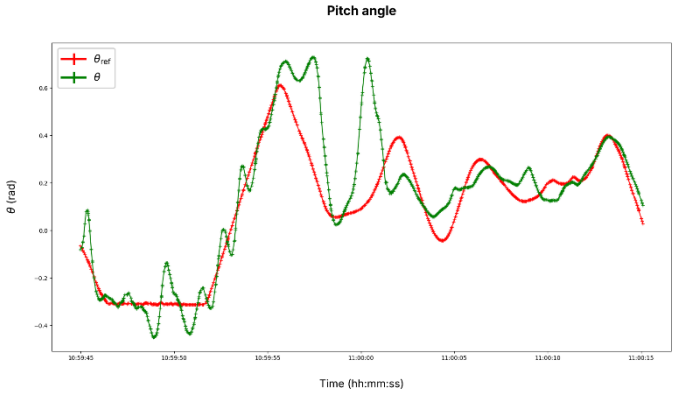


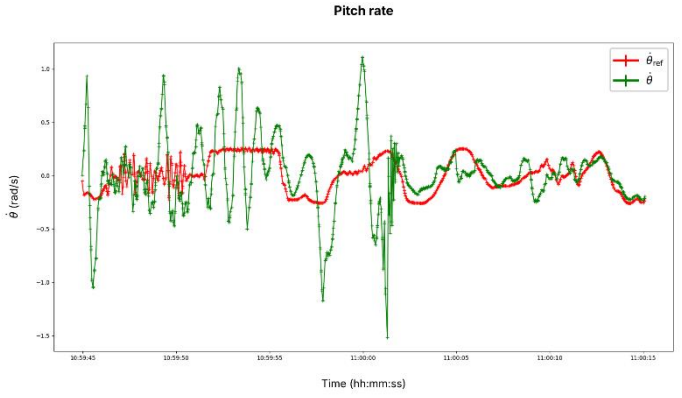


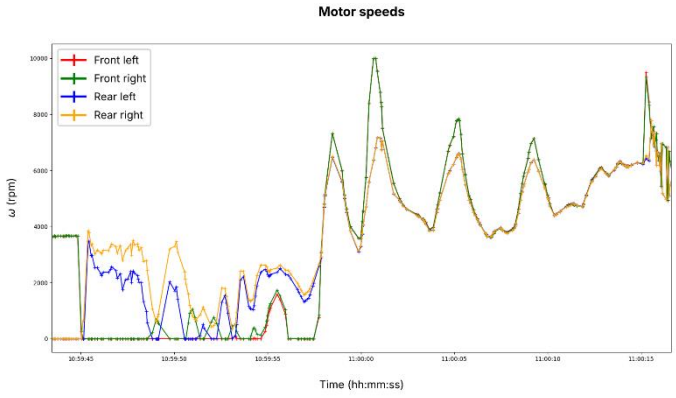


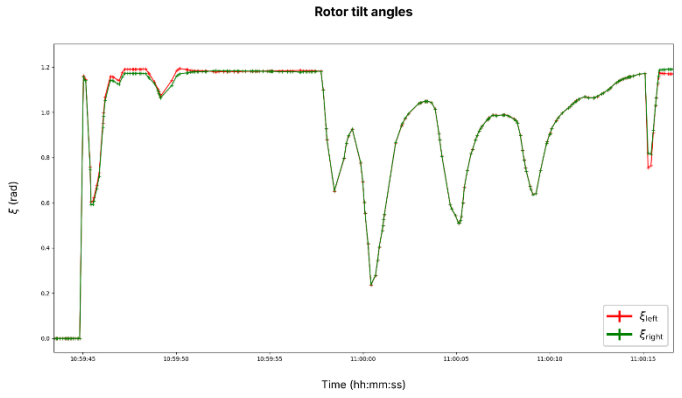
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Control – Results (cont.)









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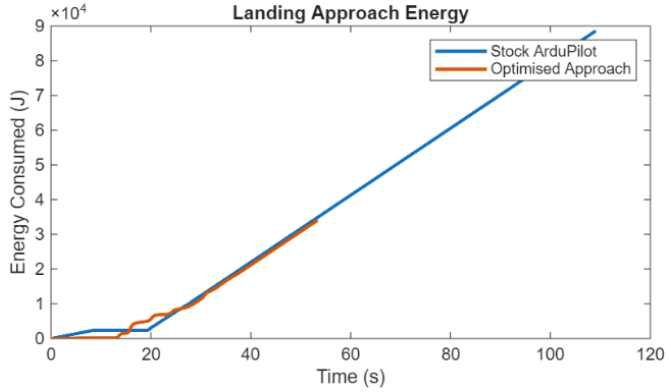
Efficiency Results

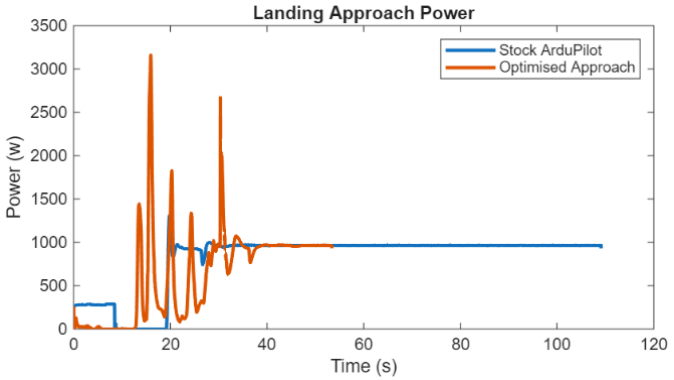
61%

less energy consumed than the stock ArduPilot landing

72%

shorter landing time





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Future Work





1. Real-world flight testing
2. Extend the model to 3D
3. Support more airframes
4. Incorporate system identification

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Thank you

Contact us

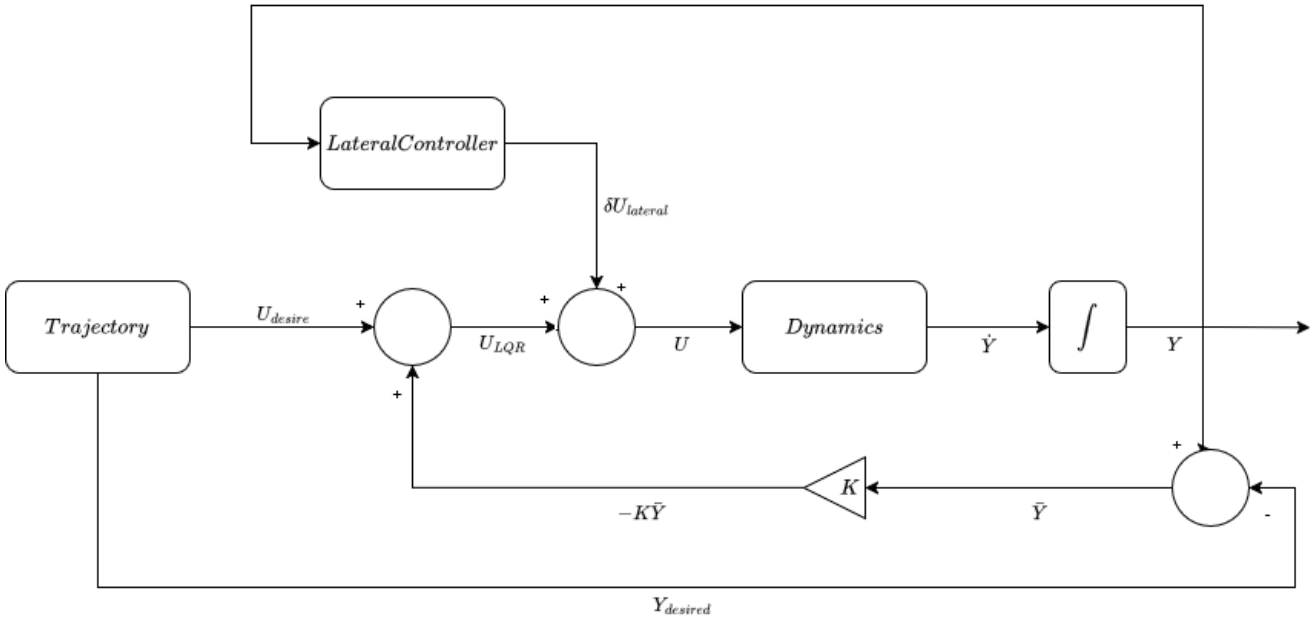
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Appendix

Control – Longitudinal



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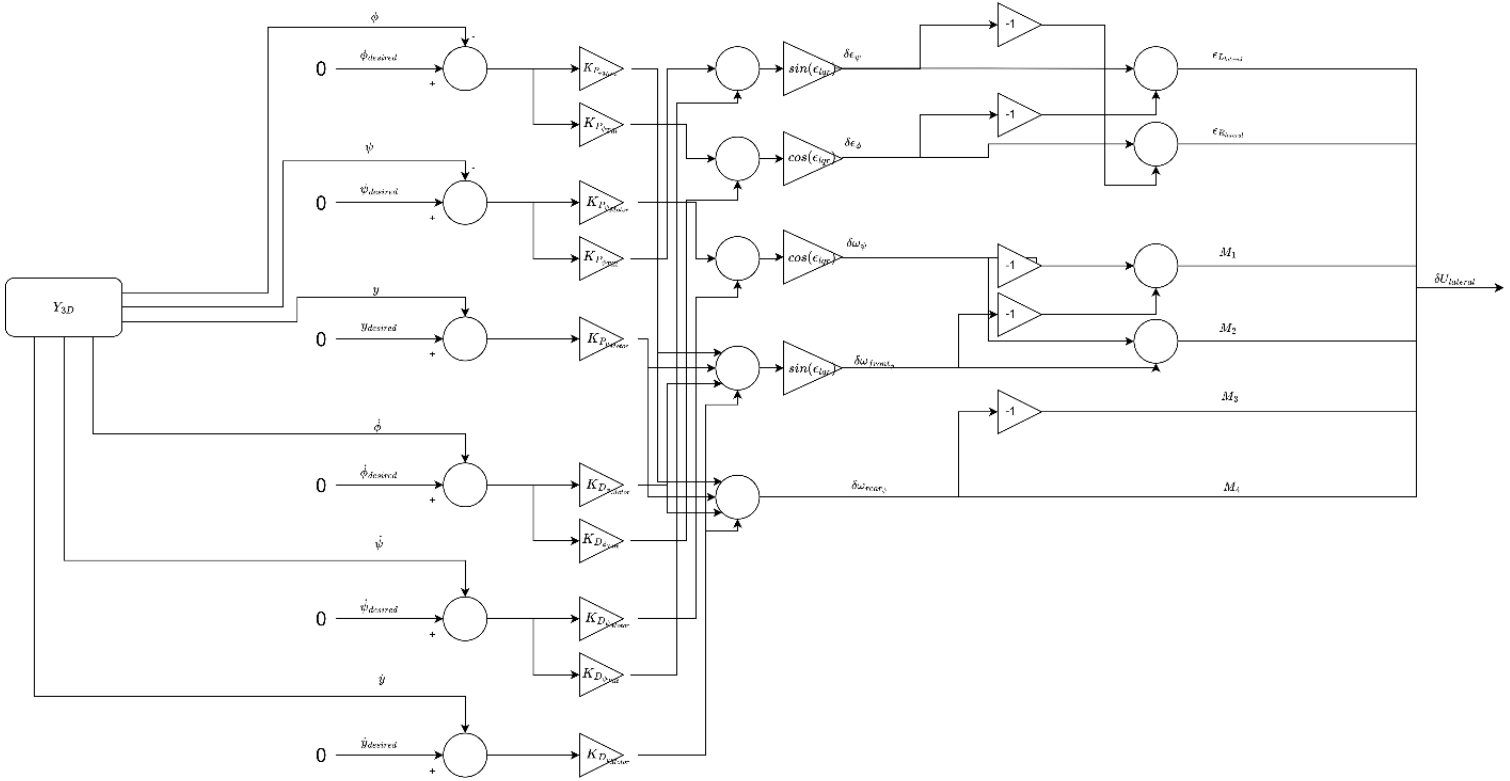
Control – Longitudinal Cost

$$J_{LQR} = \sum_{k=0}^{\frac{t_f}{\Delta t}} y_k^T Q y_k + u_k^T R u_k$$

$$Q_{ii} = \frac{1}{(max(\bar{y}_{i_{acceptable}}))^2}, R_{ii} = \frac{1}{(max(\bar{u}_{i_{acceptable}}))^2}$$

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Control – Lateral



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Rotor Power Model

$$L_r = 2\rho\pi R_r^2 v_z^i \sqrt{(v_x^i + V_{h_r})^2 + (v_z^i - V_{v_r})^2},$$

$$D_r = 2\rho\pi R_r^2 v_x^i \sqrt{(v_x^i + V_{h_r})^2 + (v_z^i - V_{v_r})^2},$$

$$P_r = L_r(v_z^i - V_{v_r}) + D_r(v_x^i + V_{h_r})$$

$$P_r = \Lambda\bar{\omega}_r^2((1+\mu_r^2)v_z^i + \mu_r V_{h_r} - V_{v_r}),$$

$$K = \frac{N_b c R_r C_T}{8\pi}, \qquad \Lambda = \frac{1}{4} N_b \rho c R_r^3 C_T.$$

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